Wk 3 Supervision Problems 1

Luca Bollini

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Group Theory

1. For each of the following groups $(G, \star)$ and subsets $H \subset G$, determine whether or not $(H, \star)$ is a subgroup. Justify your answers.

   (i) $G = 3\mathbb{Z}$, $\star =$ addition, $H = 9\mathbb{Z}$

   (ii) $G = \mathbb{R} \setminus \{0\}$, $\star =$ multiplication, $H = \{2^n : n \in \mathbb{Z}\}$

   (iii) $G = \mathbb{R}^2$, $\star =$ vector addition, $H = \{(x, y)^T : x - y \in \mathbb{Z}\}$

2. For each of the following groups, G, determine the order of the element g and write down the cyclic subgroup generated by it, $\langle g \rangle$.

   (i) $G = \mathbb{Z}/7\mathbb{Z}, g = [2]_7$

   (ii) $G = \mathbb{Z}/12\mathbb{Z}, g = [8]_{12}$

   (iii) $G = \mathrm{GL}_2(\mathbb{R})$, $g = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$

   (iv) $G = \mathrm{GL}_2(\mathbb{R})$, $g = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$

3. Prove that any group with a prime number of elements is cyclic.